

PROJECT CHARTER

Supply Chain Exception Control System

A transparent, auditable operating system for purchase-order delay and stockout exceptions

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Core stack	Python SQLite/PostgreSQL Streamlit Power BI GitHub Actions

Charter intent
This document defines the business problem, operating model, controls, success measures, and phased delivery boundary before any application code is written. All volumes, baselines, and targets are proposed simulation assumptions unless explicitly stated otherwise.

DECISION SUPPORT, NOT AUTONOMOUS CONTROL

Every exception must be explainable. Every human action must be traceable. AI may recommend, but a person remains accountable.

A. One-page condensed charter

Project purpose

Build a realistic portfolio system that converts purchase-order, inventory, demand, and supplier-performance data into prioritised operational exceptions. The system will show why each exception exists, assign an accountable owner, record actions and approvals, and provide management visibility without removing human control.

Charter area	Condensed definition
Business context	Simulated mid-sized distributor/manufacturer with 2 sites, about 1,500 open PO lines, 1,000 active SKUs, and 120 suppliers. Staff currently inspect spreadsheets and email suppliers manually. These are design assumptions, not real company facts.
Problem	Review effort is spread across all open orders; urgency is inconsistent; risks may be identified after the response window has narrowed; ownership is unclear; and actions, overrides, and closure evidence are not held in one audit trail.
Core users	Buyer/procurement analyst; inventory planner; supply-chain manager. Supporting stakeholders: warehouse/operations, finance, data/IT, internal audit, supplier account contacts, and portfolio reviewers.
In scope	Synthetic data; validation; line-level rule score; exception queue; reason codes; priority and owner; action/status history; approvals and overrides; resolution evidence; Power BI extracts and dashboard; scheduled batch; later controlled AI recommendations.
Out of scope	Live ERP integration, automatic supplier contact, autonomous expediting/cancellation, production planning optimisation, transport routing, full enterprise security, and claims of achieved commercial benefit.
Operating flow	Validate data -> score each open PO line -> create/update an explainable exception -> rank work -> assign owner -> record action/approval -> verify resolution -> report outcomes and control performance.
Success and boundary	Core success means a reviewer can trace source -> score -> action -> resolution -> BI, while no AI executes operational changes. Target: about 10 weeks at 8-10 hours/week; optional automation and AI add about 3 weeks.

Proposed acceptance target	Measure
Manual screening	From assumed 120 min/day to <=35 min/day
Critical/high ownership	>=95% assigned within 4 business hours
Explainability	100% with reason, observed value, threshold, and score contribution
Audit completeness	100% of status/priority/owner/override changes timestamped and attributed
Detection quality	>=70% recall for simulated adverse outcomes; <=35% false-positive rate for high/critical alerts
Processing	50,000 PO lines validated and scored in <=5 minutes in the test environment

Document map

Section	Content
A	One-page condensed charter
B.1-B.5	Title, context, problem, purpose, and objectives
B.6-B.8	Users, current state, and future state
B.9-B.12	Scope, exclusions, capabilities, and benefits
B.13-B.16	KPIs, assumptions, constraints, and risks
B.17-B.20	Deliverables, phases, success definition, and public statement
C	Decisions still to confirm
D	Quality self-review

Reading note

The one-page charter is suitable for a first stakeholder discussion. The detailed charter is the governing baseline for solution design. The final two sections deliberately expose unresolved decisions and weaknesses so the portfolio does not overstate certainty.

B. Detailed charter

1. Project title

Supply Chain Exception Control System (SCECS)

Working subtitle: Explainable purchase-order delay and stockout risk control for procurement operations.

2. Background and business context

Procurement and inventory teams often manage hundreds or thousands of open purchase-order lines across suppliers, sites, need dates, and stock positions. The operational problem is not a lack of data; it is the difficulty of identifying which small subset of orders requires action now and proving what was done.

This portfolio uses a simulated mid-sized distributor/manufacturer. The model assumes two receiving sites, around 1,000 active SKUs, 120 suppliers, approximately 300 purchase orders raised per week, and about 1,500 PO lines open on a typical day. Buyers use an ERP extract, inventory snapshot, demand/consumption data, supplier delivery history, spreadsheets, and email. These figures are proposed modelling assumptions only.

The system is designed as an operational control layer beside an ERP, not as a replacement for purchasing, inventory, or finance systems. It focuses attention, ownership, action tracking, and evidence. Management reporting is separated from the operational queue so each interface serves a clear purpose.

3. Problem statement

The simulated organisation cannot consistently identify, prioritise, assign, and resolve purchase-order lines that may arrive late or cause a stockout. Analysts spend time reviewing many normal orders, while risk signals are distributed across promised dates, missing confirmations, supplier reliability, projected inventory, and item criticality.

The current method produces five control gaps:

- Coverage gap: checks depend on individual spreadsheet routines and may omit orders, sites, or risk signals.
- Timing gap: risk may be discovered after the final practical expedite, substitution, or replanning window.
- Priority gap: teams may treat the oldest order as the most urgent even when another line has the greater stockout impact.
- Accountability gap: ownership and expected next action are spread across emails and personal notes.
- Evidence gap: management cannot reliably reconstruct why an alert existed, who changed it, or whether the intervention worked.

Testable problem framing

Under the assumed baseline, buyers spend about 120 minutes per working day screening open orders. The project will test whether a controlled queue can reduce screening to 35 minutes or less while retaining at least 70% recall of simulated adverse outcomes and a complete audit history.

4. Project purpose

Create an achievable, portfolio-grade exception control system that demonstrates business analysis, data modelling, ETL, rule design, operational workflow, auditability, automation, and BI. The product will reduce undirected checking by converting data into a ranked, explainable worklist while requiring people to approve and own high-impact decisions.

5. Business objectives

ID	Objective	Proposed measurable outcome
O1	Focus operational effort	Reduce assumed daily screening effort from 120 minutes to <=35 minutes without losing the agreed minimum risk coverage.

ID	Objective	Proposed measurable outcome
O2	Detect earlier	Surface high/critical simulated risks at least 2 business days before the need date where the input data makes early detection possible.
O3	Standardise priority	Apply one documented scorecard and severity matrix to 100% of eligible open PO lines.
O4	Create ownership	Assign >=95% of high/critical exceptions to a named role or user within 4 business hours of creation.
O5	Preserve control evidence	Record 100% of owner, priority, status, approval, override, and resolution events with actor, timestamp, old/new value, and reason.
O6	Enable management learning	Report exception ageing, intervention, resolution, supplier pattern, and control-quality metrics through a daily refreshed Power BI model.
O7	Govern AI safely	If the optional AI phase is built, require human approval for every recommendation that could affect an order, supplier, owner, priority, or exception status.

6. Primary users and stakeholders

Stakeholder	Role	Primary need / responsibility
Buyer / procurement analyst	Primary operator	Review assigned exceptions; contact supplier; record action, commitment, and evidence; request closure or override.
Inventory planner	Primary operator	Assess stock coverage and demand impact; propose substitution, allocation, or replanning; validate stockout resolution.
Supply-chain manager	Accountable owner	Approve rules and thresholds; reassign work; approve material overrides; monitor SLA and systemic supplier issues.
Warehouse / operations	Consulted	Confirm receipts, urgent handling constraints, and site impact.
Finance / commercial	Consulted	Advise on premium freight, order value, working capital, and supplier-commercial implications.
Data / IT owner	Enabler	Own source definitions, processing reliability, access, backups, and deployment patterns in a real implementation.
Internal audit / risk	Assurance	Review control design, evidence completeness, overrides, retention, and segregation of duties.

Stakeholder	Role	Primary need / responsibility
Supplier contact	External participant	Provides confirmation, revised date, cause, and recovery commitment; no direct system access in the MVP.
Portfolio reviewer / recruiter	Evidence consumer	Evaluates problem definition, traceability, solution trade-offs, and technical execution.

7. Current-state process

Step	Activity	Current practice	Control weakness
1	Export	Buyer exports open PO data and may separately obtain inventory and supplier reports.	Different timing, keys, and definitions; stale snapshots.
2	Filter	Buyer filters for overdue or near-due orders in a spreadsheet.	Rules vary by user; stock impact and reliability may be missed.
3	Investigate	Buyer checks emails, confirmations, stock, forecasts, and supplier history.	High context-switching and duplicated investigation.
4	Prioritise	User chooses who to chase based on experience and perceived urgency.	No consistent severity, economic impact, or reason hierarchy.
5	Act	Buyer emails/calls supplier and records notes in email or spreadsheet.	Owner, next action, commitment date, and evidence are fragmented.
6	Escalate	Serious issues are escalated through messages or meetings.	Escalation criteria and response SLA are inconsistent.
7	Report	Status is consolidated manually for meetings.	Reporting is lagging; interventions and overrides are hard to reconstruct.

8. Proposed future-state process

Future step	Operational control
1. Ingest	Scheduled batch loads PO lines, receipts, inventory, demand, item, and supplier-performance data into a dated processing run.
2. Validate	Critical keys, dates, quantities, duplicates, referential integrity, and freshness are checked. Failed critical controls stop scoring; non-critical defects are quarantined.
3. Score	A transparent rules engine calculates a 0-100 score for every eligible open PO line and stores each signal, value, threshold, points, rule version, and run ID.
4. Create/update	The engine opens, updates, suppresses, or resolves candidate exceptions according to documented lifecycle rules and deduplication keys.
5. Prioritise	The queue ranks exceptions by severity, need date, estimated stockout date, item criticality, and time since last action; users can filter but cannot erase history.

Future step	Operational control
6. Assign	An owner is allocated by buyer group, supplier, site, or manager assignment. Unassigned high/critical exceptions breach a 4-business-hour SLA.
7. Act and approve	Users record action type, note, supplier commitment, next-action date, and evidence. Material overrides require a reason and manager approval.
8. Resolve	Closure requires a resolution code and evidence such as receipt, revised demand, approved substitution, cancellation, or accepted residual risk.
9. Learn	Power BI reports backlog, ageing, SLA, outcomes, supplier patterns, false positives, overrides, and data/control health.
10. Recommend later	Optional AI summarises context or drafts recommended next steps. A user reviews, edits, approves, or rejects; the original recommendation and decision are logged.

Illustrative initial risk scorecard

The values below are proposed starting rules for simulation and calibration, not final business policy. Multiple triggered signals can contribute, capped at 100. Correlated rules must be tested to avoid double-counting.

Rule	Trigger	Points	Explanation
R01	Past confirmed delivery date and not fully received	+30	Delay already materialising
R02	Projected stockout before expected receipt	+25	Direct service/production exposure
R03	Need date within 7 days and no supplier confirmation/ASN	+15	Low visibility near response deadline
R04	Supplier trailing OTIF below 85% with sufficient history	+15	Persistent reliability signal
R05	Expected lead time exceeds item-supplier baseline by >20%	+10	Unusual lead-time variance
R06	Item marked operationally critical	+5	Raises consequence, not likelihood
M01	Approved substitute or protected stock covers need	-15	Documented mitigation reduces consequence

Score	Severity	Proposed response
0-29	Monitor	No operational exception; retain score history for analysis.
30-49	Medium	Queue for review; owner within 1 business day.
50-69	High	Owner within 4 business hours; action due within 8 business hours.
70-100	Critical	Immediate queue priority; manager visibility; action plan required within 4 business hours.

9. Project scope

CORE SCOPE

- A documented simulated operating context, data dictionary, process map, user stories, acceptance criteria, and control matrix.
- Synthetic historical and current-state datasets covering suppliers, items, sites, PO headers/lines, receipts, inventory snapshots, demand/consumption, confirmations, and supplier performance.
- Python-based generation, ingestion, validation, transformation, score calculation, and test evidence.
- SQLite for the local MVP, with data access patterns and migration notes that permit PostgreSQL later without redesigning the business model.
- A line-level exception lifecycle with stable exception ID, source/run lineage, reason breakdown, severity, owner, status, SLA dates, actions, approvals, overrides, evidence references, and resolution code.
- A Streamlit operational interface for work queues, exception detail, action capture, assignment, approval/override, and resolution demonstration.
- Power BI semantic model and management pages for control health, operational backlog, supplier performance, ageing, outcomes, and intervention analysis.
- A scheduled batch through GitHub Actions for validation, scoring, test execution, and publishable data extracts; persistence limits will be documented.
- Optional notifications and optional AI recommendations only after the core controls pass acceptance tests.

DATA AND WORKFLOW BOUNDARY

The primary grain is one open purchase-order line at one processing run. One persistent operational exception represents a PO-line risk episode and may span multiple runs. Every recalculation is retained in score history; every user mutation is retained in event history. Duplicate alert creation is prevented through a documented exception key and lifecycle rule.

10. Out-of-scope items

Excluded area	Boundary
Production integration	Direct SAP/Oracle/Dynamics integration, EDI, supplier portals, warehouse systems, and real-time event streaming.
Autonomous execution	Automatic order change, cancellation, expedite purchase, premium freight approval, supplier sanction, inventory allocation, or user reassignment.
Advanced optimisation	Network optimisation, safety-stock optimisation, production scheduling, transport routing, or prescriptive sourcing.
Enterprise platform controls	Full SSO, enterprise RBAC, segregation-of-duties engine, disaster recovery, penetration testing, 24/7 support, and regulated records management.
External supplier access	Supplier login, direct message delivery, contractual workflow, or performance dispute resolution.
Claimed realised benefits	No claim that cost, service level, or labour savings were achieved in a real organisation; targets are portfolio acceptance hypotheses.
AI in the MVP	No autonomous decision-making and no AI dependency for scoring, alert creation, assignment, approval, or closure.

11. Main system capabilities

ID	Capability	Operational requirement
C01	Synthetic data and scenario control	Generate reproducible normal, delay, demand-spike, missing-data, supplier-deterioration, and mitigation scenarios using independent random seeds.

ID	Capability	Operational requirement
C02	Data validation and quarantine	Check schema, types, keys, dates, quantities, duplicates, freshness, and relationships; stop critical failures and isolate non-critical defects.
C03	Explainable risk engine	Calculate score by versioned rules and persist rule ID, reason text, actual value, threshold, points, and data/run lineage.
C04	Exception lifecycle	Create, update, suppress, reopen, and resolve line-level exceptions without duplicating an active risk episode.
C05	Priority work queue	Sort/filter by severity, site, supplier, buyer, need date, stockout date, age, and action due date.
C06	Ownership and SLA	Assign accountable owner and track assignment/action/closure SLAs, including unassigned and overdue queues.
C07	Action and evidence log	Record action, note, commitment, next step, evidence reference, actor, and time in append-only history.
C08	Approval and override control	Require reason and authority for material score/severity/status exceptions; retain original and approved values.
C09	Resolution and outcome	Close with structured resolution/cause codes and evidence; link later receipt/stockout outcome to the risk episode.
C10	Management BI	Provide daily snapshots and measures for backlog, ageing, SLA, risk reasons, suppliers, interventions, overrides, outcomes, and data health.
C11	Scheduled processing	Run validation, scoring, tests, and export on a schedule with failure status and run metadata.
C12	Controlled AI support - optional	Summarise exception context and draft next-step options; require review and log prompt context, model/version, output, user decision, and final action.

12. Expected business benefits

Benefit	Mechanism	Proposed evidence / target
Focused labour	Screen only material exceptions rather than every open order.	Proposed reduction from 120 to <=35 screening minutes per buyer/day; time study in usability test.
Earlier response	Expose risk while expedite, substitution, reallocation, or demand discussion is still possible.	>=70% of correctly detected high/critical simulated outcomes first flagged >=2 business days before need date, where the source signal exists.
Consistent priority	Use one score and response matrix instead of individual judgement alone.	100% of eligible lines scored under a stored rule version; overrides reported separately.
Visible accountability	Make owner, due action, and escalation status explicit.	>=95% high/critical assigned within 4 business hours; <10% open actions overdue after stabilisation.

Benefit	Mechanism	Proposed evidence / target
Auditability	Reconstruct the decision trail for challenge, learning, and governance.	100% required state changes attributed and timestamped; no destructive history edits through the UI.
Supplier insight	Distinguish recurring supplier patterns from one-off operational noise.	Trend exceptions, OTIF, root cause, recovery commitment, and repeat-risk rate by supplier/item/site.
Safer AI adoption	Demonstrate AI value without delegating high-impact control.	100% of AI recommendations reviewed; 0 direct AI order/status/priority changes.

13. Success metrics and KPIs

All targets below are proposed portfolio acceptance targets. They are not achieved results and should not be presented as employer or client outcomes until measured in a real pilot.

KPI	Definition	Proposed target	Measurement
Control coverage	% eligible open PO lines scored per successful run	>=99.5%	Eligible scored / eligible received
Explainability completeness	% exceptions containing reason, actual, threshold, points, rule version, run ID	100%	Complete exceptions / exceptions
Critical data validity	% records passing mandatory key/date/quantity checks	>=99.5%; critical failure stops run	Valid / tested records
Batch reliability	% scheduled runs completed successfully	>=98% over 30 simulated runs	Successful / scheduled runs
Batch performance	Elapsed validation and scoring time for 50,000 PO lines	<=5 minutes	Run end - run start
High/critical recall	% simulated adverse outcomes preceded by high/critical alert	>=70%	True positives / all adverse outcomes
High/critical false-positive rate	% high/critical alerts with no adverse outcome within outcome window	<=35%	False positives / high+critical alerts
Useful lead time	Median business days from first high/critical alert to need date	>=2 days	Need date - first alert date
Assignment SLA	% high/critical exceptions assigned in 4 business hours	>=95%	Assigned in SLA / new high+critical
Action SLA	% high/critical exceptions with action plan in 8 business hours	>=90%	Actioned in SLA / new high+critical
Overdue action rate	% open actions past next-action due date	<10%	Overdue open actions / open actions
Resolution cycle time	Median hours from open to verified resolution by severity	Baseline first; target after 3 test cycles	Resolved timestamp - created timestamp
Manual screening time	Average active minutes required to identify daily priority work	<=35 minutes vs assumed 120	Timed scenario test
Audit completeness	% controlled state changes with actor, timestamp, before/after, reason	100%	Complete events / controlled events
Dashboard freshness	Age of latest successful dataset shown in Power BI	<=26 hours for daily batch	Now - latest run time
AI human review - optional	% AI recommendations approved/rejected before operational use	100%; 0 autonomous changes	Reviewed / generated recommendations

14. Key assumptions

ID	Assumption
A01	Each PO line has a stable identifier and can be linked to item, supplier, site, ordered quantity, received quantity, order date, and promised/expected date.
A02	Inventory and demand data are available at item-site level often enough to estimate projected coverage.
A03	Supplier delivery history is sufficient to calculate a meaningful reliability indicator only after a minimum observation threshold; otherwise the rule is marked unavailable, not zero-risk.
A04	The simulated business accepts line-level exception ownership and the proposed severity/SLA concept for demonstration.
A05	Users can record actions and resolution evidence without changing the source ERP transaction in the MVP.
A06	Business hours, holidays, time zone, currency, and unit-of-measure rules can be simplified for the first release but will be explicit.
A07	Power BI consumes curated extracts or PostgreSQL tables; direct SQLite connectivity is not assumed for the final demonstration.
A08	GitHub Actions is used for reproducible scheduled processing and tests, not as an enterprise job scheduler or durable operational database.
A09	The portfolio repository contains only synthetic data and no confidential employer, customer, or supplier information.
A10	The core project is completed before notifications or AI work begins.

15. Constraints

Constraint	Design implication
Single developer	One early-career analyst must design, build, test, document, and demonstrate the system; prioritisation and modular delivery are essential.
Synthetic evidence	No real operational users, source systems, supplier communications, or cost outcomes; usability and business targets are simulated acceptance tests.
Technology fit	SQLite is simple and portable but limited for concurrent writes and enterprise access; PostgreSQL adds setup and hosting complexity.
Streamlit control limits	Streamlit can demonstrate workflow but is not, by itself, a full enterprise case-management or access-control platform.
Power BI portability	A PBIX can be built and documented, but cloud refresh/sharing may depend on licences, gateway, and hosting not included in the portfolio.
Scheduled persistence	GitHub-hosted runners are ephemeral; durable operational state must not depend on a runner-local SQLite file.

Constraint	Design implication
Data realism	Demand, lead-time, confirmations, and disruptions must be simulated without making the risk score trivially predict labels generated by the same logic.
Time budget	Core MVP target: about 10 weeks at 8-10 hours/week. Notifications and AI are optional, not prerequisites for success.

16. Risks and mitigations

ID	Risk	Exposure	Mitigation / control
R1	Synthetic outcomes are circular because the same rules generate labels and predict them.	High	Generate operational events and outcomes with separate stochastic logic/seeds; retain hidden outcome labels; test on unseen scenarios.
R2	Correlated signals double-count risk and inflate severity.	High	Document rule dependencies; inspect contribution patterns; cap categories; calibrate thresholds against outcome and workload.
R3	Too many alerts recreate the manual workload.	High	Set a daily queue capacity assumption; measure precision and alerts per 100 lines; introduce suppression, cooldown, and deduplication rules.
R4	A missed critical order creates false confidence.	High	Report recall and missed-event review; include data-quality warnings; never describe score as a guarantee; keep user search and manual escalation.
R5	Missing/stale data produces misleading low risk.	High	Treat missing critical signals as data exceptions; show freshness; stop or qualify scoring rather than defaulting to safe.
R6	SQLite or Streamlit is presented as enterprise-ready.	Medium	Label the MVP boundary; demonstrate architecture separation; document PostgreSQL, authentication, deployment, concurrency, and backup needs.
R7	Audit history can be edited or lost.	High	Use append-only event patterns, stable IDs, timestamps, actor context, automated integrity tests, backups/exports, and UI restrictions.
R8	GitHub Actions overwrites or loses operational state.	High	Use Actions for reproducible batch/test/export only; write durable state to an approved persistent store or commit versioned synthetic outputs where appropriate.
R9	Power BI metrics disagree with operational UI.	Medium	Define shared business rules and run IDs; reconcile record counts and severity totals; publish freshness and failed-run indicators.

ID	Risk	Exposure	Mitigation / control
R10	Users override scores to clear workload without resolving risk.	Medium	Require structured reason, evidence, expiry, and manager approval for material override; report override rate and outcomes.
R11	AI generates plausible but unsafe recommendations.	High	Use retrieval from approved fields only; restrict permitted output; require human review; log output and decision; prohibit tool execution and autonomous mutation.
R12	Project scope expands beyond one analyst's capacity.	High	Gate phases with acceptance criteria; make notifications and AI optional; prefer one complete end-to-end workflow over broad integrations.

17. High-level deliverables

ID	Deliverable	Completion evidence
D01	Business analysis pack	Charter, problem statement, stakeholder map, current/future process, scope, business rules, user stories, acceptance criteria, decisions and risk log.
D02	Data design pack	Conceptual/logical model, source-to-target mapping, data dictionary, lineage, data-quality rules, test scenarios, and synthetic data specification.
D03	Risk-control specification	Eligibility, rule catalogue, score bands, rule versioning, reason format, deduplication, suppression, escalation, override, resolution, and calibration method.
D04	Reproducible data pipeline	Python generation/ETL/validation/scoring workflow, configuration, run metadata, tests, failure handling, and sample outputs.
D05	Operational MVP	Streamlit queue, exception detail, assignment, action, approval/override, resolution, filters, and audit-history views.
D06	Analytics layer	Curated star schema/extracts, KPI definitions, reconciliation tests, and Power BI dashboard pages.
D07	Automation pack	GitHub Actions schedule, test gates, run artefacts/log status, failure notification design, and persistence boundary.
D08	Governance pack	Access/role concept, audit model, data retention, model/rule change control, AI guardrails, and known limitations.
D09	Portfolio evidence	README, architecture diagram, screenshots, demo script, test results, KPI evidence, design decisions, limitations, and future roadmap.
D10	Optional AI proof of concept	Context summary/next-action recommendation with structured output, approval/rejection flow, audit fields, evaluation, and kill switch.

18. Proposed project phases

Phase	Indicative timing	Scope	Exit criterion
0. Charter and design	Week 1	Confirm scenario, users, data grain, rules, KPI definitions, scope, and acceptance plan.	Charter approved; unresolved decisions logged.
1. Data foundation	Weeks 2-3	Design schema; generate independent scenarios; validate data; establish lineage and quality report.	Reproducible dataset; critical tests pass; data dictionary complete.
2. Risk engine	Weeks 4-5	Implement eligibility, rule contributions, bands, reason codes, run history, outcomes, and backtest/calibration.	Every alert explainable; accuracy/workload tests meet agreed threshold or limitations documented.
3. Operational MVP	Weeks 6-8	Build queue, details, assignment, actions, status, approvals/overrides, resolution, and audit history.	End-to-end scenario trace passes; no destructive history update through normal UI.
4. Power BI	Weeks 9-10	Create analytics model, measures, reconciliation, management pages, and refresh/freshness indicators.	Counts reconcile to operational store; KPI pages answer agreed management questions.
5. Automation - optional	Week 11	Schedule validation/scoring/export/tests; document persistence and failure-handling boundary.	Repeatable scheduled run and visible failure outcome.
6. Controlled AI - optional	Weeks 12-13	Add summary/next-step recommendation, human decision, audit record, evaluation, and kill switch.	0 autonomous mutations; 100% recommendations reviewed in test scenarios.

Recommended release strategy

Treat Phase 4 as the core portfolio finish line. Phases 5 and 6 should be added only if the end-to-end exception lifecycle, audit trail, testing, and documentation are already credible. A complete controlled workflow is stronger evidence than a partially integrated AI feature.

19. Definition of project success

The project is successful when the core release can be demonstrated and independently reviewed against all of the following conditions:

- A reproducible synthetic scenario contains normal orders, late receipts, missing confirmations, demand shocks, supplier deterioration, mitigations, data defects, and outcome labels generated independently from the score rules.
- A single run validates and scores all eligible lines, stops on critical data failure, and stores run metadata, rule version, and record-level lineage.
- Every operational exception displays a plain-language explanation with source value, threshold, points, and total score; a reviewer can reproduce the total.
- A user can assign, act, escalate, override with required approval, resolve with evidence, and view the complete history without overwriting earlier events.
- Operational and Power BI totals reconcile by run, date, severity, status, supplier, and site, with freshness and failed-run visibility.
- Acceptance evidence reports workload, recall, false positives, lead time, SLAs, audit completeness, and known limitations; targets that fail are disclosed rather than hidden.
- The repository contains setup guidance, architecture, data definitions, tests, decisions, screenshots, demo flow, governance controls, and a clear distinction between simulated targets and achieved results.

- If AI is included, it cannot create, close, reprioritise, reassign, approve, or execute an operational decision; every output and human decision is auditable.

20. Concise project statement for GitHub and LinkedIn

GITHUB README STATEMENT

Supply Chain Exception Control System

A portfolio-grade procurement control system that uses Python and a transparent rule engine to identify purchase-order lines at risk of delay or stockout, explain every risk signal, prioritise operational work, and maintain an auditable record of ownership, actions, overrides, approvals, and resolution. Streamlit supports the operational workflow, Power BI provides management monitoring, and optional AI recommendations remain subject to human review.

LINKEDIN PROJECT STATEMENT

Portfolio project

I am designing an end-to-end Supply Chain Exception Control System for a simulated procurement operation. The project combines business analysis, synthetic data engineering, explainable risk rules, a Streamlit action queue, audit controls, scheduled processing, and Power BI reporting. The core principle is controlled decision support: every exception shows why it exists, and people remain responsible for approvals, overrides, and outcomes. All performance figures will be reported as simulated test results, not real client savings.

C. Unclear decisions still to confirm

These decisions should be resolved before detailed design or explicitly retained as documented assumptions. Recommended defaults are included to keep the project moving without hiding uncertainty.

ID	Decision	Question	Recommended default
D1	Business scenario	Distributor, manufacturer, or hybrid? What is the operational consequence of stockout?	Hybrid distributor/manufacturer with two sites; define both customer-service and production-critical items.
D2	Dataset scale	How many suppliers, SKUs, sites, years, PO lines, and daily open lines?	2 sites, 120 suppliers, 1,000 SKUs, 24 months, 50,000+ historical lines, about 1,500 open lines/day.
D3	Primary exception grain	PO header, PO line, item-site, or supplier episode?	PO line for action; aggregate item-site and supplier risk for reporting.
D4	Delay definition	Use promised date, confirmed date, requested need date, or receipt date? How are partial receipts treated?	Evaluate unreceived quantity against latest accepted promised date and operational need date; document partial receipt logic.
D5	Stockout method	Simple days-of-cover, projected available balance, or forecast distribution?	Start with deterministic projected available balance; show assumptions and add probabilistic modelling only later.
D6	Demand source	Forecast, sales orders, average consumption, production requirement, or combination?	Use simulated daily demand/requirements with forecast error and event shocks.
D7	Criticality	Who defines item criticality and can users override it?	Master-data flag with manager-approved, time-bound override; criticality affects consequence only.
D8	Score calibration	What daily alert volume and miss tolerance is operationally acceptable?	Assume a buyer capacity of 15-25 new exceptions/day and prioritise recall for critical items.
D9	Ownership	Assign by buyer, supplier, commodity, site, or manual manager decision?	Default buyer-group mapping; allow audited manager reassignment.
D10	Lifecycle	Exact status names, reopen rules, suppression window, and closure evidence?	New, Assigned, Investigating, Action Agreed, Monitoring, Resolved, Closed; suppress duplicates while active and reopen on material deterioration.
D11	Approvals	Which overrides require manager approval and what is the authority threshold?	Any severity reduction, accepted residual risk, or closure without receipt/substitution requires approval.
D12	Business calendar	Working hours, weekends, holidays, and time zone for SLA calculations?	Australia/Melbourne business calendar; 08:30-17:00 weekdays; explicit holiday table.
D13	Database	SQLite-only demonstration or PostgreSQL-backed deployed application?	SQLite for local core; PostgreSQL only after workflow is stable or if multi-user deployment is demonstrated.
D14	Power BI connection	Direct database, exported CSV/Parquet, or API?	Curated CSV/Parquet for portable demo; PostgreSQL views for an optional hosted architecture.

ID	Decision	Question	Recommended default
D15	Deployment	Local demo, Streamlit Community Cloud, or another hosted environment?	Local demo first; public hosting only with synthetic data and explicit limitations.
D16	Authentication	Is simulated role switching sufficient, or is real login required?	Simulated roles for the portfolio; clearly state that it is not production security.
D17	Notification channel	Email, Teams/Slack, or in-app only?	In-app overdue queue first; add one simulated notification channel only after core success.
D18	AI use case	Summary, recommended action, root-cause classification, or message draft?	Start with context summary and 2-3 constrained action options; no free-form automation.
D19	AI data boundary	Which fields may be sent to a model and what retention/privacy policy applies?	Synthetic, minimised, approved fields only; store model/version, input reference, output, and review decision.
D20	Public evidence	Will PBIX, data, screenshots, and demo be public?	Publish synthetic sample data, documentation, screenshots, and reproducible tests; keep secrets and tokens outside the repository.

D. Quality self-review

Overall assessment: the charter is credible as a portfolio operating-control design, but it should not be presented as a production solution or evidence of real business impact. The strongest elements are explainability, lifecycle ownership, measurable acceptance criteria, and explicit AI governance. The weakest areas are data realism, validation of thresholds, and deployment architecture.

Weakness / realism issue	Impact	Critical analysis	Required treatment
Synthetic benefits cannot be validated commercially	High	The time, recall, false-positive, and SLA figures are hypotheses. They demonstrate test discipline but do not prove savings or service improvement.	Label every figure as proposed or simulated; report measured test results separately; avoid ROI claims without real cost and baseline data.
Risk score may appear more precise than it is	High	A 0-100 score can imply statistical probability even though it is a prioritisation index.	Call it a risk priority score, not probability. Show points and bands; do not interpret 70 as a 70% delay chance.
Outcome simulation may be circular	High	If outcome labels use the same thresholds as the score, performance metrics become meaningless.	Generate events/outcomes independently, use unseen seeds/scenarios, and include noise, missingness, recovery actions, and concept drift.
Stockout logic is simplified	High	Real stockout exposure depends on forecast error, allocations, safety stock, substitute rules, production constraints, and receipt uncertainty.	Use deterministic projected balance for MVP, state limitations, and avoid claiming optimised inventory decisions.
Score signals can overlap	High	Overdue, near need date, missing confirmation, and lead-time variance may describe the same underlying problem.	Test contribution correlation, cap categories, conduct sensitivity analysis, and calibrate against workload and missed outcomes.
Targets lack a real baseline	Medium	The assumed 120-minute screening time and response SLAs are not observed.	Treat them as acceptance-test parameters. If possible, run a small timed usability session and publish the protocol.
SQLite and Streamlit are not enterprise controls	High	Concurrent transactions, access control, immutable logs, backups, and deployment security are limited.	Present them as an MVP; isolate data/service/UI layers; document PostgreSQL, authentication, row-level access, logging, and backup migration.
Power BI and SQLite fit is weak	Medium	Direct connectivity is inconvenient and cloud refresh may require additional components.	Use curated CSV/Parquet for the portable demo or PostgreSQL views for a hosted architecture; reconcile outputs by run ID.
GitHub Actions is not durable operational hosting	High	Runner storage is temporary and schedules are not guaranteed like enterprise orchestration.	Use it for tests and reproducible batch/export; keep state in a durable store; disclose scheduling limitations.
Auditability is demonstrated, not guaranteed	High	An append-only UI does not prevent database administrators or code from changing rows.	Use event tables, database constraints where practical, checksums/export, backups, and role separation in the target architecture; phrase claims carefully.
Supplier action remains outside the system	Medium	The system records contact but cannot verify email/call content or supplier commitment automatically.	Require structured commitment fields/evidence reference and treat communication integration as out of scope.

Weakness / realism issue	Impact	Critical analysis	Required treatment
AI phase may distract from stronger core evidence	Medium	A rushed chatbot would weaken the governance story and consume limited build time.	Make AI optional and begin only after data quality, risk rules, workflow, audit, tests, and BI pass exit criteria.
One-person schedule is ambitious	Medium	A 10-week core requires disciplined scope and may be disrupted by study/work commitments.	Use vertical slices, weekly exit criteria, sample-size limits, and a firm core finish line; remove hosted deployment before cutting controls or tests.
No real user validation	Medium	The workflow could be technically complete but operationally awkward.	Ask 1-3 procurement/supply-chain practitioners to review screenshots or a scripted prototype; record feedback as design evidence, not endorsement.

Final quality judgement

Proceed with conditions

The project is appropriate for one early-career analyst if the core release ends at a complete, tested exception workflow plus Power BI and if deployment, notifications, and AI remain optional. The portfolio should be judged by traceability, controls, evidence, and honest limitations - not by the number of technologies included.